

AYUSH PATEL

Software Engineer | Machine Learning & LLM Applications

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SUMMARY

Computer Science graduate (B.S., 2025) who builds full-stack and ML-driven applications with React, Python, Node.js, and Java. Strengths in prompt engineering and LLM application development, REST API design, and shipping projects end-to-end.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, HTML/CSS, C#

Concepts: Data Structures & Algorithms, Object-Oriented Programming, REST APIs, SDLC, Agile

Frameworks & Libraries: React.js, Node.js, Express.js, .NET, Tailwind CSS, Framer Motion, PyTorch, OpenCV, pandas, NumPy
AI / ML: Machine learning, deep learning, LLM fine-tuning (LoRA/QLoRA), Hugging Face, prompt engineering, LLM application development

Tools & Databases: Git/GitHub, GitHub Pages, GitHub Actions (CI/CD), Vite, AWS, Microsoft Azure, SQL Server, MongoDB, VS Code, Google Colab, Figma

TECHNICAL PROJECTS

PyExplain — AI Python Code Explainer

Jun 2026

LLM fine-tuning | Python, PyTorch, Hugging Face PEFT, Qwen2.5-Coder-7B, LoRA/QLoRA | [Model](#) • [GitHub](#)

- Fine-tuned Qwen2.5-Coder-7B with QLoRA on a curated dataset of 56 Python code-explanation examples to explain code in plain English for users with zero programming background, publishing the LoRA adapter to Hugging Face.
- Iterated from a local 1.5B prototype to the 7B model trained on a free Google Colab GPU for simpler, more accurate explanations, with a one-click Colab demo and reproducible training notebook; runs locally in 4-bit (~4.5 GB).

Personal Portfolio Website

Aug 2025 - Present

Deployed React site | React, Vite, Tailwind CSS, Framer Motion, GitHub Pages | [Live](#) • [GitHub](#)

- Designed and built a responsive personal portfolio in React and Tailwind CSS, with animated transitions via Framer Motion and interactive particle effects.
- Deployed the site to GitHub Pages through an automated Vite build, showcasing projects and contact information at ayushpatel2803.github.io.

Enchanted Wars — Senior Design

Jan 2025 - Apr 2025

Browser-based card game | React.js, Node.js, Express.js, Git | [GitHub](#)

- Independently built the entire responsive frontend in React with reusable components, state management, and UI animations for a smooth cross-device experience.
- Integrated the React frontend with the Node.js/Express REST API for game-state sync and matchmaking, set up a GitHub Actions CI workflow, and handled testing and project presentations through a mid-project client withdrawal.

RSNA 2024 Lumbar Spine Degenerative Classification

Jan 2024 - Apr 2024

Medical image classification | Python, OpenCV, Google Colab, Agile | [Colab](#)

- Supported a 5-person Agile team building a Python/OpenCV model to classify lumbar-spine MRI degeneration, owning testing, documentation, and progress logging, and presenting results.

WORK EXPERIENCE

Amazon — Warehouse Associate

Plymouth, MI

Apr 2023 - Jul 2024

- Exceeded daily fulfillment targets with high pick accuracy in a high-volume operation, coordinating with team members under shifting priorities and safety standards.

CVS Health & Pharmacy — Customer Associate

Plymouth, MI

Sep 2021 - Oct 2022

- Resolved customer issues and onboarded new team members while consistently meeting daily store service goals.

EDUCATION

University of Michigan-Dearborn

Dearborn, MI

B.S. in Computer Science | GPA: 3.3

Aug 2021 - Apr 2025

Relevant coursework: Algorithms & Data Structures, Operating Systems, Web Development, Senior Design